RAJALAKSHMI ENGINEERING COLLEGE

Harikarasudhan S 251001070-2025-IT

H2

| | |
|---|---|
| **Started on** | Thursday, 6 August 2026, 11:08 AM |
| **State** | Finished |
| **Completed on** | Saturday, 10 October 2026, 11:50 PM |
| **Time taken** | 65 days 12 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>
int main() {
    int n,rev=0,remainder;
    int counter=0;
    scanf("%d",&n);
    counter++;
    while(n!=0)
    {
        counter++;
        remainder=n%10;
        counter++;

        rev=rev*10 + remainder;
        counter++;

        n/=10;
        counter++;
    }
    counter++;
    counter++;
    printf("%d",counter);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course